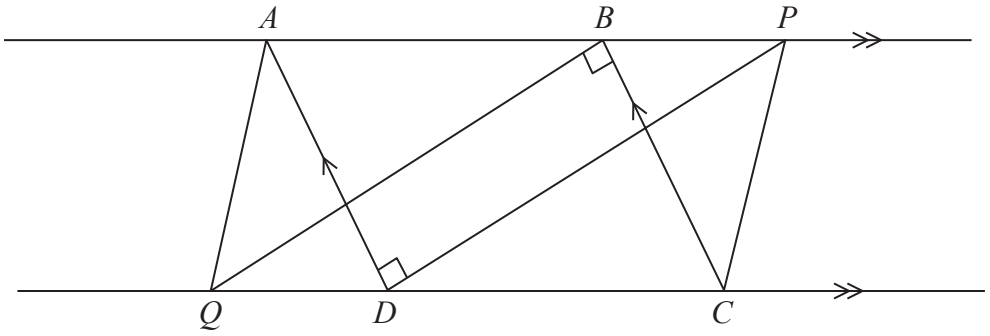
21



ABCD is a parallelogram.
ABP and *QDC* are straight lines.
Angle *ADP* = angle *CBQ* = 90°

(a) Prove that triangle *ADP* is congruent to triangle *CBQ*.

AD = BC: opposite sides of a parallelogram are equal.

1 mark
First equal feature, with its reason

Angle PAD = angle QCB: opposite angles of parallelogram ABCD.

Angle ADP = angle CBQ = 90° (given).

1 mark
Both equal angles and the included equal side, with reasons

Therefore triangle ADP is congruent to triangle CBQ by ASA.

1 mark
State the congruence test

(3)

(b) Explain why *AQ* is parallel to *PC*.

AP = CQ, because corresponding sides of congruent triangles are equal.

1 mark
Use the congruence to identify equal opposite sides

Also AP is parallel to CQ: AB and DC lie on parallel straight lines.

One pair of opposite sides of APCQ is both equal and parallel,

so APCQ is a parallelogram. Therefore AQ is parallel to PC.

(2)

1 mark
Identify the parallelogram, then use its opposite sides

(Total for Question 21 is 5 marks)